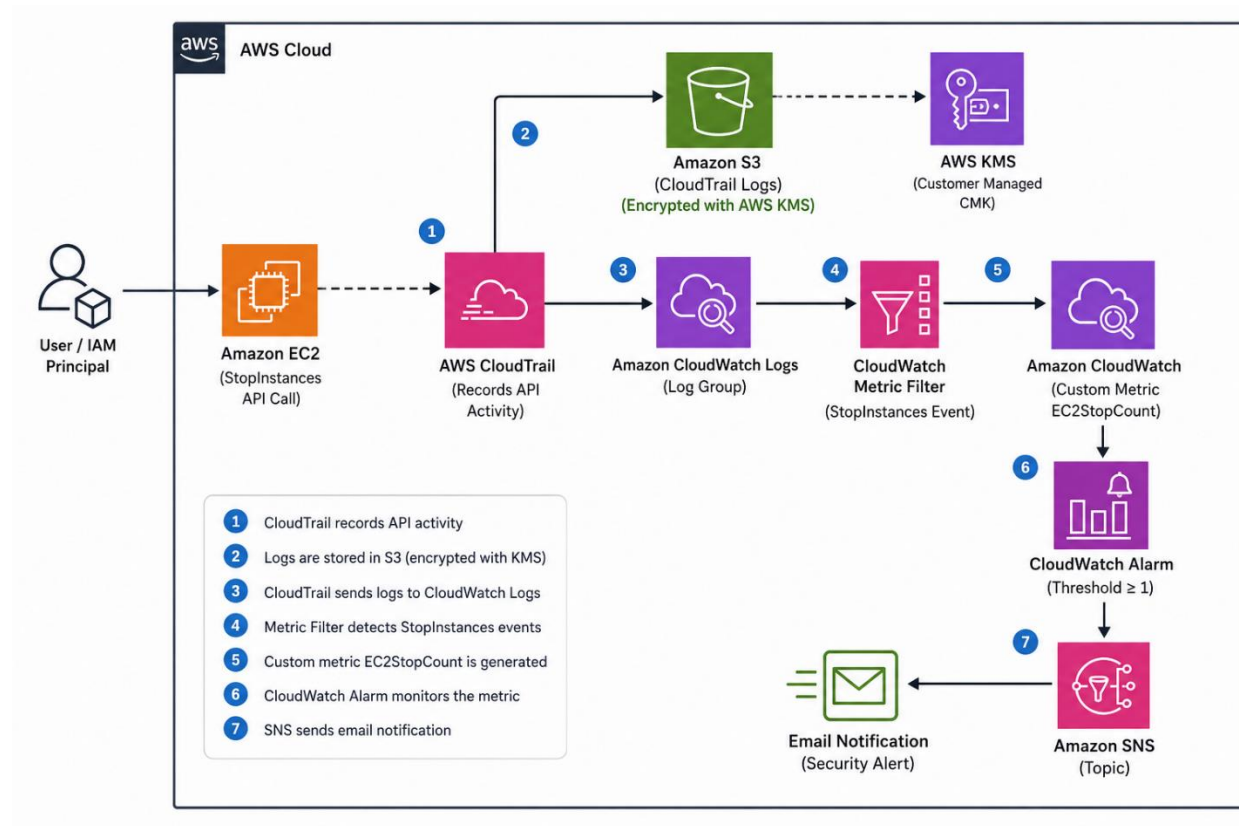


AWS CloudTrail Security Monitoring Project

Project Overview

This project demonstrates how to build a centralized AWS security monitoring solution using AWS services. AWS CloudTrail records AWS management API activities, securely stores audit logs in an encrypted Amazon S3 bucket using AWS Key Management Service (KMS), and streams log events to Amazon CloudWatch Logs. CloudWatch Metric Filters detect EC2 StopInstances API activity, generate custom metrics, and trigger CloudWatch Alarms based on security conditions. Amazon SNS then delivers real-time email notifications when suspicious or monitored activities are detected.

This project demonstrates the practical implementation of AWS logging, monitoring, encryption, and automated alerting to improve visibility and security event detection within an AWS environment.



Project Objectives

- Monitor AWS management API activities using AWS CloudTrail.
- Securely store CloudTrail logs in Amazon S3.
- Encrypt audit logs using AWS KMS.
- Detect EC2 StopInstances API events.
- Generate CloudWatch custom metrics.
- Trigger CloudWatch Alarms based on security events.
- Send automated email notifications using Amazon SNS.

AWS Services Used

Service	Purpose
AWS CloudTrail	Records AWS API activities
Amazon S3	Stores CloudTrail logs
AWS KMS	Encrypts CloudTrail log files
Amazon CloudWatch Logs	Receives CloudTrail log events
CloudWatch Metric Filters	Detects StopInstances events
CloudWatch Alarms	Monitors custom metrics
Amazon SNS	Sends email notifications
Amazon EC2	Test resource
AWS IAM	Provides permissions for CloudTrail

Implementation Steps

Step 1 – Create an Amazon S3 Bucket

Created an S3 bucket to store CloudTrail log files.

Configuration:

- Block Public Access enabled
- Bucket Versioning enabled
- Default encryption enabled (AWS KMS)

Purpose:

- Secure storage of audit logs
- Prevent accidental deletion using versioning

Create bucket [Info](#)

Buckets are containers for data stored in S3.

General configuration

AWS Region

Europe (Stockholm) eu-north-1

Bucket type [Info](#)

☒ General purpose

Recommended for most use cases and access patterns. General purpose buckets are the original S3 bucket type. They allow a mix of storage classes that redundantly store objects across multiple Availability Zones.

☐ Directory

Recommended for low-latency use cases. These buckets use only the S3 Express One Zone storage class, which provides faster processing of data within a single Availability Zone.

Bucket namespace

Choose the namespace where you want to create your bucket. [Learn more](#)

☒ Global namespace

By default, S3 creates general purpose buckets in the global namespace.

☐ Account Regional namespace (recommended)

General purpose buckets created in your account Regional namespace are unique to your account. These buckets can never be created by another AWS account.

Bucket name [Info](#)

store-events456

Bucket names must be 3 to 63 characters and unique within the global namespace. Bucket names must also begin and end with a letter or number. Valid characters are a-z, 0-9, periods (.), and hyphens (-). [Learn more](#)

Block Public Access settings for this bucket

Public access is granted to buckets and objects through access control lists (ACLs), bucket policies, access point policies, or all. In order to ensure that public access to this bucket and its objects is blocked, turn on Block all public access. These settings apply only to this bucket and its access points. AWS recommends that you turn on Block all public access, but before applying any of these settings, ensure that your applications will work correctly without public access. If you require some level of public access to this bucket or objects within, you can customize the individual settings below to suit your specific storage use cases. [Learn more](#)

☒ Block all public access

Turning this setting on is the same as turning on all four settings below. Each of the following settings are independent of one another.

☒ Block public access to buckets and objects granted through new access control lists (ACLs)

S3 will block public access permissions applied to newly added buckets or objects, and prevent the creation of new public access ACLs for existing buckets and objects. This setting doesn't change any existing permissions that allow public access to S3 resources using ACLs.

☒ Block public access to buckets and objects granted through any access control lists (ACLs)

S3 will ignore all ACLs that grant public access to buckets and objects.

☒ Block public access to buckets and objects granted through new public bucket or access point policies

S3 will block new bucket and access point policies that grant public access to buckets and objects. This setting doesn't change any existing policies that allow public access to S3 resources.

☒ Block public and cross-account access to buckets and objects through any public bucket or access point policies

S3 will ignore public and cross-account access for buckets or access points with policies that grant public access to buckets and objects.

Default encryption [Info](#)

Server-side encryption is automatically applied to new objects stored in this bucket.

Encryption type [Info](#)

Secure your objects with two separate layers of encryption. For details on pricing, see [DSSE-KMS pricing](#) on the [Storage](#) tab of the [Amazon S3 pricing page](#).

☐ Server-side encryption with Amazon S3 managed keys (SSE-S3)

☒ Server-side encryption with AWS Key Management Service keys (SSE-KMS)

☐ Dual-layer server-side encryption with AWS Key Management Service keys (DSSE-KMS)

AWS KMS key [Info](#)

☒ Choose from your AWS KMS keys

☐ Enter AWS KMS key ARN

Available AWS KMS keys

arn:aws:kms:eu-north-1:319627299994:key/e9ddad0c-c0ff-4305-9c22-9ae65ba546d3



Create a KMS key [Info](#)

Bucket Versioning

Versioning is a means of keeping multiple variants of an object in the same bucket. You can use versioning to preserve, retrieve, and restore every version of every object stored in your Amazon S3 bucket. With versioning, you can easily recover from both unintended user actions and application failures. [Learn more](#)

Bucket Versioning

- ☐ Disable
- ☒ Enable

Step 2 – Create an AWS KMS Key

Created a customer-managed KMS key to encrypt CloudTrail log files.

Configuration:

- Symmetric encryption key
- IAM administrator assigned
- Key users configured
- Updated KMS Key Policy to allow CloudTrail to use the key

Purpose:

- Encrypt CloudTrail log files
- Improve data protection
- Meet security best practices

Configure key

Key type [Help me choose](#)

- ☒ **Symmetric**
A single key used for encrypting and decrypting data or generating and verifying HMAC codes

- ☐ **Asymmetric**
A public and private key pair used for encrypting and decrypting data, signing and verifying messages, or deriving shared secrets

Key usage [Help me choose](#)

- ☒ **Encrypt and decrypt**
Use the key only to encrypt and decrypt data.

- ☐ **Generate and verify MAC**
Use the key only to generate and verify hash-based message authentication codes (HMAC).

► Advanced options

Add labels

Alias

You can change the alias at any time. [Learn more](#)

Alias

cloudtrail-kmskey

Define key administrative permissions - optional

Key administrators (1/18)

Select the IAM users and roles authorized to manage this key via the KMS API. These administrators will be added to the key policy under the statement identifier (Sid) 'Allow administration of the key'. Modifying this Sid might impact the console's ability to update the administrator statement in the key policy. [Learn more](#)

Q Search Key administrators				< 1 2 >
☐	Name	Path	Type	
☒	jigs	/	User	

Key users (1/18)

Select the IAM users and roles authorized to use this key in cryptographic operations. These users will be added to the key policy under the statement identifiers (Sid) 'Allow use of the key' and 'Allow attachment of persistent resources'. Modifying these Sids might impact the console's ability to update the user statements in the key policy. [Learn more](#)

Q Search Key users				< 1 2 >
☐	Name	Path	Type	
☒	jigs	/	User	

Key policy

```
71     },
72     {
73       "Sid": "Allow CloudTrail to use the KMS key",
74       "Effect": "Allow",
75       "Principal": {
76         "Service": "cloudtrail.amazonaws.com"
77       },
78       "Action": [
79         "kms:GenerateDataKey*",
80         "kms:DescribeKey",
81         "kms:Encrypt"
82       ],
83       "Resource": "*"
84     }
85   ]
86 }
```

Customer managed keys (2)

Filter keys by properties or tags

Key ID = e9ddad0c-c0ff-4305-9c22-9ae65ba546d3

Clear filter

1 matches

<1>

☐	Aliases	Key ID	Status	Key type	Key spec	Key usage
☐	cloudtrail-...	e9ddad0c-c0ff-4305-9c22-9ae65ba546d3	Enabled	Symmetric	SYMMETR...	Encrypt and decrypt

Step 3 – Create an Amazon SNS Topic

Created an SNS topic for security alerts.

Configuration:

- Protocol: Email
- Subscription endpoint: Email address

Purpose:

- Send automatic email notifications when alarms are triggered.

Create topic

Details

Type [Info](#)
Topic type cannot be modified after topic is created

☐ FIFO (first-in, first-out)

- Strictly-preserved message ordering
- Exactly-once message delivery
- Subscription protocols: SQS

☒ Standard

- Best-effort message ordering
- At-least once message delivery
- Subscription protocols: SQS, Lambda, Data Firehose, HTTP, SMS, email, mobile application endpoints

Name

eventsnotify01

Maximum 256 characters. Can include alphanumeric characters, hyphens (-) and underscores ().

Topic eventsnotify01 created successfully.
You can create subscriptions and send messages to them from this topic.

[Publish message](#) ×

Notifications 0 0 1 1 0 0

eventsnotify01 [Edit](#) [Delete](#) [Publish message](#)

Details

Name	ARN	Display name	Type
eventsnotify01	arn:aws:sns:eu-north-1:319627299994:eventsnotify01	-	Standard

Topic owner
319627299994

[Subscriptions](#) [Access policy](#) [Delivery policy \(HTTP/S\)](#) [Delivery status logging](#) [Encryption](#) [Tags](#) [Integrations](#)

Subscriptions (0) [Edit](#) [Delete](#) [Request confirmation](#) [Confirm subscription](#) [Create subscription](#)

Search

ID	Endpoint	Status	Protocol
----	----------	--------	----------

Create subscription

Details

Topic ARN

Protocol
The type of endpoint to subscribe

Email

Endpoint
An email address that can receive notifications from Amazon SNS.

Step 4 – Create an AWS CloudTrail Trail

Created a multi-region CloudTrail.

Configuration:

- Existing encrypted S3 bucket
- Customer-managed KMS key
- Management Events:
 - Read
 - Write
- Log File Validation enabled
- CloudWatch Logs integration enabled
- IAM role created for CloudTrail

Purpose:

- Record AWS API activity
- Deliver logs to Amazon S3
- Stream logs to CloudWatch Logs

- Step 1
● Choose trail attributes
- Step 2
○ Choose log events
- Step 3
○ Review and create

Choose trail attributes

General details

A trail created in the console is a multi-region trail. [Learn more](#)

Trail name

Enter a display name for your trail.

events-tracker

3-128 characters. Only letters, numbers, periods, underscores, and dashes are allowed.

☐ Enable for all accounts in my organization

To review accounts in your organization, open AWS Organizations. [See all accounts](#)

Storage location [Info](#)

☐ Create new S3 bucket

Create a bucket to store logs for the trail.

☒ Use existing S3 bucket

Choose an existing bucket to store logs for this trail.

Trail log bucket name

Enter a new S3 bucket name and folder (prefix) to store your logs. Bucket names must be globally unique.

store-events456

X

Browse

Prefix - optional

Log file SSE-KMS encryption [Info](#)

☒ Enabled

Customer managed AWS KMS key

☐ New

☒ Existing

AWS KMS alias

arn:aws:kms:eu-north-1:319627299994:key/e9ddad0c-c0ff-4305-9c22-9ae65ba546d3

X

▼ Additional settings

Log file validation [Info](#)

☒ Enabled

SNS notification delivery [Info](#)

☐ Enabled

CloudWatch Logs - optional

Configure CloudWatch Logs to monitor your trail logs and notify you when specific activity occurs. Standard CloudWatch and CloudWatch Logs charges apply. [Learn more](#)

CloudWatch Logs [Info](#)

☒ Enabled

Log group [Info](#)

☒ New

☐ Existing

Log group name

cloudtrail/loggroup

1-512 characters. Only letters, numbers, dashes, underscores, forward slashes, and periods are allowed.

IAM Role [Info](#)

AWS CloudTrail assumes this role to send CloudTrail events to your CloudWatch Logs log group.

☒ New

☐ Existing

Role name

Cloudtrailassumerole_events-tracker

Events [Info](#)

Record API activity for individual resources, or for all current and future resources in AWS account. [Additional charges apply](#)

Event type

Choose the type of events that you want to log.

☒ **Management events**

Capture management operations performed on your AWS resources.

☐ **Data events**

Log the resource operations performed on or within a resource.

☐ **Insights events**

Identify unusual activity, errors, or user behavior in your account.

☐ **Network activity events**

Network activity events provide information about resource operations performed on a resource within a virtual private cloud endpoint.

Management events [Info](#)

Management events show information about management operations performed on resources in your AWS account.

i No additional charges apply to log management events on this trail because this is your first copy of management events.

API activity

Choose the activities you want to log.

☒ **Read**

☒ **Write**

☐ **Exclude AWS KMS events**

☐ **Exclude Amazon RDS Data API events**

Step 5 – CloudWatch Log Group

Log groups | Data sources - new | Summary

► Get-set-go with unified data store

Log groups (2) [Refresh](#) [Filters](#) [Actions](#) [View in Logs Insights](#) [Start tailing](#) [Create log group](#)

By default, we only load up to 10,000 log groups.

Q cloudtrail/loggroup X 1 match ☒ Exact match

☐	Log group	Log class	Anomaly d...	Deleti...	Bearer...	Dat...	Sen...	Retenti...
☐	cloudtrail/loggroup	Standard	Configure	Off	Off	-	-	Never exp...

Step 6 – Create CloudWatch Metric Filter

Inside the CloudWatch Log Group:

Created a Metric Filter to detect EC2 StopInstances events.

Example Filter Pattern

```
{ $.eventName = "StopInstances" }
```

Define pattern

Create filter pattern

You can use metric filters to monitor events in a log group as they are sent to CloudWatch Logs. You can monitor and count specific terms or extract values from log events and associate the results with a metric. [Learn more about pattern syntax.](#)

Filter pattern

Specify the terms or pattern to match in your log events to create metrics.

Q { \$.eventName = "StopInstances" } X

☐ Enable metric filter on transformed logs

Metric Configuration

- Namespace: CloudTrailSecurity1
- Metric Name: EC2StopCount
- Metric Value: 1

Purpose:

- Convert CloudTrail log events into CloudWatch metrics.

Create filter name

Log events that match the pattern you define are recorded to the metric that you specify. You can graph the metric and set alarms to notify you.

Filter name

EC2StopInstancesFilter

Filter pattern

{ \$.eventName = "StopInstances" }

☐ Enable metric filter on transformed logs
When enabled, metric filter will be applied to transformed logs. When disabled, metric filter will be applied to original logs.

Metric details

Metric namespace

Namespaces let you group similar metrics. [Learn more](#)

CloudTrailSecurity1

[Create new](#)

Namespaces can be up to 255 characters long; all characters are valid except for colon(:) at the start of the name.

Metric name

Metric name identifies this metric, and must be unique within the namespace. [Learn more](#)

EC2StopCount

Metric name can be up to 255 characters long; all characters are valid except for colon(:), asterisk(*), dollar(\$), and space().

Metric value

Metric value is the value published to the metric name when a Filter Pattern match occurs.

1

Valid metric values are: floating point number (1, 99.9, etc.), numeric field identifiers (\$1, \$2, etc.), or named field identifiers (e.g. \$requestSize for delimited filter pattern or \$.status for JSON-based filter pattern - dollar (\$) or dollar dot (\$.) followed by alphanumeric and/or underscore (.) characters).

Step 7 – Create CloudWatch Alarm

Created an alarm based on the EC2StopCount metric.

Configuration

- Statistic: Sum
- Period: 5 Minutes
- Threshold:
 - $EC2StopCount \geq 1$

- Alarm Action:
 - Send notification to Amazon SNS

Purpose:

- Detect EC2 stop operations in near real time.

Alarms (0)

Clear selection Create composite alarm Actions **Create alarm**

Filter alarms Quick filters

Name	State	Actions	Actions muted - new	Last state update (UTC)	Condition
No alarms					
No alarms to display					
Read more about Alarms					
Create alarm					

Data source
Select the type of data to monitor

☒ **Metrics**
Monitor CloudWatch metrics

☐ **Logs**
Monitor CloudWatch log queries

☐ **Alarms**
Combine multiple alarms

Type
Choose the metric type for this alarm.

☒ **Classic**
Use the standard CloudWatch metric selection experience.

☐ **PromQL**
Use PromQL queries to define the metric for this alarm.

Metric
Graph
Preview of the metric or metric expression and the alarm threshold.

[Select metric](#)

[Cancel](#) [Next](#)

[Browse \(1,694\)](#) Multi source query Graphed metrics Options Source

[Add math](#) [Add query](#)

☐ Alarm recommendations

[Graph with SQL](#) [Graph search](#)

Search for any metric, dimension, resource id or account id

▼ Custom namespaces

CloudTrailSecurity1	1
---------------------	---

Browse (1)

Multi source query

Graphed metrics

Options

Source

All > CloudTrailSecurity1

Q Search for any metric, dimension, resource id or account id

Metrics with no dimensions

1

Browse (1)

Multi source query

Graphed metrics (1)

Options

Source

Add math

Add query

All > CloudTrailSecurity1 > Metrics with no dimensions

Alarm recommendations

Graph with SQL

Graph search

Q Search for any metric, dimension, resource id or account id

< 1 >

✓ Metric name 1/1

▲ Alarms

✓ EC2StopCount

No alarms

Cancel

Select metric

CloudWatch > Alarms > Create alarm

0 14:00 14:30 15:00 15:30 16:00 16:30

EC2StopCount

Statistic

Q Sum

Period

5 minutes

Conditions

Threshold type

☒ Static
Use a value as a threshold

☐ Anomaly detection
Use a band as a threshold

Whenever EC2StopCount is...

☐ Greater
> threshold

☒ Greater/Equal
≥ threshold

☐ Lower/Equal
≤ threshold

☐ Lower
< threshold

than...
Define the threshold value.

1

Notification

Alarm state trigger
Define the alarm state that will trigger this action. Remove

☒ **In alarm**
The metric or expression is outside of the defined threshold.

☐ **OK**
The metric or expression is within the defined threshold.

☐ **Insufficient data**
The alarm has just started or not enough data is available.

Send a notification to the following SNS topic
Define the SNS (Simple Notification Service) topic that will receive the notification.

☒ **Select an existing SNS topic**

☐ Create new topic

☐ Use topic ARN to notify other accounts

Send a notification to...

Only topics belonging to this account are listed here. All persons and applications subscribed to the selected topic will receive notifications.

Email (endpoints)
lucife48476@gmail.com - [View in SNS Console](#)

Add notification

Step 8 – Test the Solution

Created an EC2 instance.

Instance state = running		Clear filters		< 1 >			
☒	Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Publ
☒	server1	i-0882547f604904cd6	Running	t3.micro	Initializing	eu-north-1c	ec2-

Stopped the instance after creating the Metric Filter.

Instances (1/1) [Info](#)

Last updated
7 minutes ago

[Connect](#)

[Instance state ▾](#)

[Actions ▾](#)

[Launch instances](#) ▾

Saved filter sets

Running ▾

Instance state = running X

[Clear filters ▾](#)

< 1 >

☒	Name	Instance ID	Instance state ▾	Instance type ▾	Status check	Availability Zone ▾	Publ
☒	server1	i-0882547f604904cd6	⏸ Stopped	t3.micro	–	eu-north-1c	–

Expected Results:

- CloudTrail records the StopInstances API event.

Lookup attributes

Event name < 1 >

Event name	Event time	User name	Event source	Resource type	Resource name
StopInstances	July 14, 2026, 22:43:30 (UTC+0...)	jigs	ec2.amazonaws.com	AWS::EC2::Instance	i-0882547f604904cd6

1 / 5 events selected

Event record

Resource referenced 1

Event properties	Event 1
Resource type	AWS::EC2::Instance
Resource name	i-0882547f604904cd6 Link

AWS Config resource

- CloudWatch Logs receives the log.
- Metric Filter increments EC2StopCount.
- CloudWatch Alarm changes from **OK** to **ALARM**.

Alarms (1)

☐	Name	State	Actions	Actions muted - new	Last state update (UTC)	Condition
☐	EC2InstancesStop	In alarm	☒ Actions enabled	-	2026-07-14 17:00:54	EC2StopC minutes

- Amazon SNS sends an email notification.

ALARM: "EC2InstancesStop" in EU (Stockholm)

AWS Notifications <no-reply@sns.amazonaws.com> 10:45 PM (0 minutes ago) ☆ 😊 ↶ ⋮

to me ▾

You are receiving this email because your Amazon CloudWatch Alarm "EC2InstancesStop" in the EU (Stockholm) region has entered the ALARM state, because "Threshold Crossed: 1 out of the last 1 datapoints [1.0 (14/07/26 16:55:00)] was greater than or equal to the threshold (1.0) (minimum 1 datapoint for OK -> ALARM transition)." at "Tuesday 14 July, 2026 17:00:54 UTC".

View this alarm in the AWS Management Console:
<https://eu-north-1.console.aws.amazon.com/cloudwatch/deeplink.js?region=eu-north-1#alarmsV2:alarm/EC2InstancesStop>

Alarm Details:

- Name: EC2InstancesStop
- Description:
- State Change: **INSUFFICIENT_DATA -> ALARM**
- Reason for State Change: Threshold Crossed: 1 out of the last 1 datapoints [1.0 (14/07/26 16:55:00)] was greater than or equal to the threshold (1.0) (minimum 1 datapoint for OK -> ALARM transition).
- Timestamp: Tuesday 14 July, 2026 17:00:54 UTC
- AWS Account: 319627299994
- Alarm Arn: arn:aws:cloudwatch:eu-north-1:319627299994:alarm:EC2InstancesStop

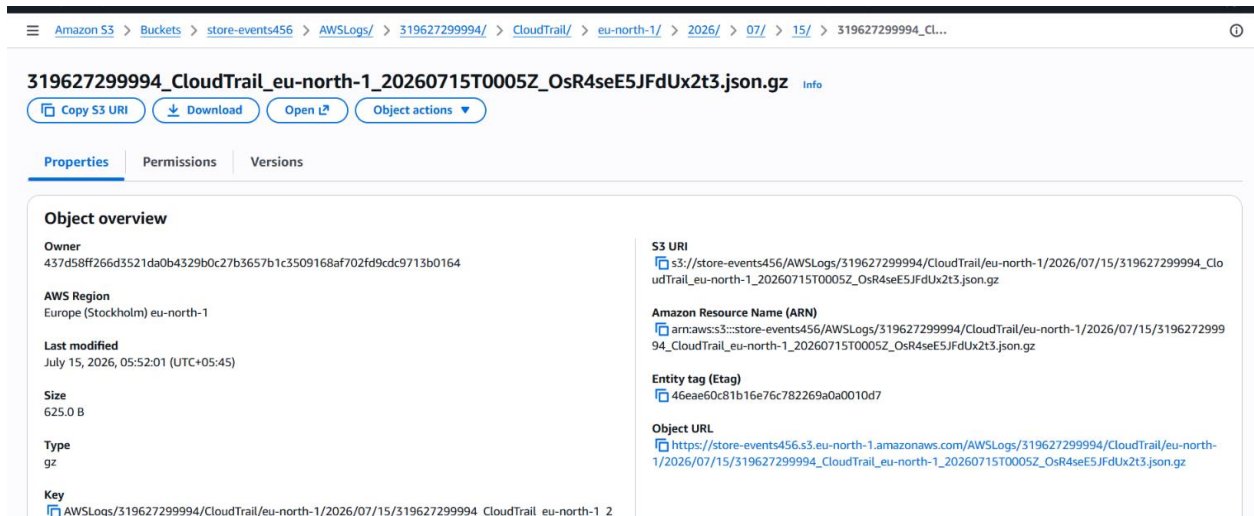
Threshold:

- The alarm is in the ALARM state when the metric is GreaterThanOrEqualToThreshold 1.0 for at least 1 of the last 1 period(s) of 300 seconds.

Monitored Metric:

- MetricNamespace: **CloudTrailSecurity1**
- MetricName: EC2StopCount

- Encrypted CloudTrail log files are stored in Amazon S3.



Results

The solution successfully achieved the following:

- Recorded AWS API activities using AWS CloudTrail.
- Stored encrypted audit logs in Amazon S3.
- Protected log files using AWS KMS.
- Streamed CloudTrail events to Amazon CloudWatch Logs.
- Detected EC2 StopInstances events using Metric Filters.
- Triggered CloudWatch Alarms automatically.
- Delivered email notifications through Amazon SNS.

Security Features

- CloudTrail enabled for API auditing
- AWS KMS encryption for log protection
- Amazon S3 versioning enabled
- CloudWatch monitoring
- Real-time alerting
- Email notifications
- Secure IAM role for CloudTrail integration

Project Outcome

Successfully built a centralized AWS security monitoring solution capable of:

- Monitoring AWS API activities
- Detecting EC2 StopInstances events
- Encrypting audit logs with AWS KMS
- Storing logs securely in Amazon S3

- Sending automated email alerts using Amazon SNS
- Improving visibility into security-related activities within an AWS environment

Challenges Faced

During the implementation, the following challenges were encountered and resolved:

- Configuring the AWS KMS key policy to allow CloudTrail encryption.
- Creating the correct IAM trust relationship for CloudTrail to publish logs to CloudWatch Logs.
- Understanding that CloudWatch Metric Filters only process new log events created after the filter is configured.
- Configuring the CloudWatch Alarm with the correct statistic, threshold, and evaluation period.

Key Learnings

- AWS CloudTrail captures AWS management API activities.
- AWS KMS protects audit logs through encryption.
- CloudWatch Metric Filters convert log events into custom metrics.
- CloudWatch Alarms provide automated monitoring based on custom metrics.
- Amazon SNS enables real-time notification delivery.

Skills Demonstrated

- AWS CloudTrail
- Amazon CloudWatch
- CloudWatch Logs
- CloudWatch Metric Filters
- CloudWatch Alarms
- Amazon SNS
- AWS KMS
- Amazon S3
- AWS IAM
- Amazon EC2
- AWS Security Monitoring
- Cloud Logging
- Incident Detection
- Event Monitoring